4 (a)	Upon release, the **elastic force** exerted by the spring **on the copper ball is larger** than the **copper ball's weight**. Hence, the direction of the **resultant force is upwards**.	**B1** **B1**
(b)	The additional elastic force experienced by the copper ball is due to the initial displacement of x_0. This force is also opposite in direction to the displacement x. $\sum F = ma$ $ma = -kx$ $a = -\frac{k}{m}x$	**C1** **M1**
(c)	The copper ball's **acceleration is directly proportional to its displacement *x* since *k* is a constant** and **the negative sign shows acceleration is always directed towards a fixed equilibrium position**.	**B1** **B1**
(d)(i)	$\Delta\omega / \omega = ½ \Delta k/k + ½ \Delta m/m$ $= ½ (0.2 / 13.6) + ½ (0.005 / 1.200)$ $= \underline{0.0094}$ (2 sf)	**M1** **A1**
(ii)	$\omega = \sqrt{\frac{13.6}{1.200}} = 3.3665$ $\Delta\omega = 0.03$ (1 sf) Hence, $\omega = \underline{3.37 \pm 0.03}$ rad s^{-1}	 **C1** **A1**
(iii)	Total energy $= ½ m \omega_0^2 x_0^2$ $= ½ (1.200) (3.37)^2 (0.20)^2$ $= 0.27257$ J $= \underline{0.27}$ J	**C1** **A1**